

# Orvyn — Comprehensive Technical Implementation Report

*Audit-and-Compliance OS for Indian Mid-Tier CA Firms and Enterprise Internal Audit Teams*

## TL;DR

- Build the 14-day demo on Next.js 15 + shadcn/ui + Drizzle + Postgres (Neon Mumbai) + Gemini 2.0 Flash + Docling + rapidfuzz + @noble/ed25519; deploy the V1 90-day production stack on AWS Mumbai (ap-south-1) with Aurora Postgres + RLS, ECS Fargate, a NestJS workspace API + FastAPI AI service split, LiteLLM self-hosted, Bedrock Claude Sonnet (Geo CRIS APAC) for high-stakes verification, and Temporal for long-running orchestration in V1.5+.
- Port the Aditya Birla verification engine as a four-agent Pydantic AI pipeline (planner → extractor → matcher → reviewer) using Docling for OCR/layout, rapidfuzz + Gemini text-embedding-004 for matching, and react-pdf-highlighter for click-to-source — targeting ≤5 minutes for a 50-page PDF vs. 20-sheet Excel at ~₹12 LLM cost per verification.
- Run a Turborepo monorepo with a hierarchical CLAUDE.md, 4-5 narrowly-scoped subagents (not generic role-based), GitHub + Linear + Postgres + Sentry MCP servers, and skills encoding CA-specific knowledge (CARO 2020, SA 230, 3CD, GSTR-9C). The

**single highest-leverage move for the 8-person team is keeping main-session context under ~40% via aggressive subagent dispatch.**

## **Key Findings**

- 1. OSS landscape favours permissive licenses for everything except OCR.** Docling (MIT, 59,593 stars on github.com/docling-project/docling as of May 12, 2026, IBM-Research-backed, hosted under LF AI & Data) is the clear winner for the document AI layer. Marker — while excellent at 30.9k stars on the canonical datalab-to/marker repo — is GPL with CC-BY-NC weights and requires a commercial license above \$2M revenue/funding. rapidfuzz is MIT (vs. fuzzywuzzy GPL). react-pdf-highlighter is MIT. Orvyn can ship V1 with zero copyleft contamination if it avoids Marker. (GitHub + 5)
- 2. Pydantic AI is the agent framework for the Verification Engine.** v1.0 shipped Sept 2025; 17,024 stars on the official Pydantic GitHub org as of May 12, 2026; its type-safety model maps cleanly onto Pydantic schemas Orvyn will already use for FastAPI. LangGraph is the right choice only when human-in-the-loop checkpointing is needed (PBC List Generator in V1.5). (Speakeasy + 2)
- 3. DPDP Act hard deadline is 13 May 2027.** All substantive obligations (consent, breach reporting, data principal rights, security safeguards, SDF requirements) must be met by that date with no grace period. India data residency for personal data (Rule 14) is operative since MeitY's notification of 13 November 2025. Design the stack around Vertex AI Gemini

in Mumbai as the default and Bedrock Claude (via Geo CRIS APAC) for high-stakes work.

DLA Piper + 3

4. **Multi-tenancy: Drizzle ORM + Postgres RLS, not Prisma.** Drizzle ships first-class RLS helpers (`crudPolicy()`), is ~200x smaller bundle than Prisma, edge-native; Prisma has no built-in RLS. For audit-tech where every query must filter by `tenant_id` and Postgres-enforced isolation is a procurement requirement, Drizzle wins. `BuildMVPFast` `BuildMVPFast`
5. **The CA-specific Indian stack is shallow but functional.** Tally integration via XML-over-HTTP (default port 9000) works but OSS Python wrappers are immature — Orvyn must write its own. Leegality is the right primary e-sign (Aadhaar eSign is a recognised electronic signature under the Second Schedule of the IT Act, 2000). Sarvam AI is V2 (client-portal Indian languages), not V1. `Tally`

## Details

---

# PART 1 — OPEN-SOURCE REPOS AND LIBRARIES TO LEVERAGE

For each entry: GitHub URL · License · Stars (May 2026 where known) · Verdict.

## A. Audit / compliance / financial workspace UIs

| Project                       | URL                                                                                 | License    | Verdict                                                                                              |
|-------------------------------|-------------------------------------------------------------------------------------|------------|------------------------------------------------------------------------------------------------------|
| <b>ERPNext</b> (audit module) | <a href="https://github.com/frappe/erpnext">github.com/frappe/erpnext</a>           | GPLv3      | Learn from only — GPL incompatible with closed SaaS. Study chart-of-accounts and ledger UI patterns. |
| <b>Akaunting</b>              | <a href="https://github.com/akaunting/akaunting">github.com/akaunting/akaunting</a> | GPLv3      | Learn from. UI patterns for invoices/expenses; do not link in.                                       |
| <b>Tremor</b>                 | <a href="https://github.com/tremorlabs/tremor">github.com/tremorlabs/tremor</a>     | Apache 2.0 | Integrate. Dashboards for the partner cockpit. Maps cleanly onto shadcn.                             |
| <b>recharts</b>               | <a href="https://github.com/recharts/recharts">github.com/recharts/recharts</a>     | MIT        | Integrate. Primary chart layer.                                                                      |

| Project                  | URL                                                                                   | License      | Verdict                                                                             |
|--------------------------|---------------------------------------------------------------------------------------|--------------|-------------------------------------------------------------------------------------|
| <b>observablehq/plot</b> | <a href="https://github.com/observablehq/plot">github.com/observablehq/plot</a>       | ISC          | Use for one-off explanatory charts only.                                            |
| <b>Outline</b>           | <a href="https://github.com/outline/outline">github.com/outline/outline</a>           | BSL → Apache | Learn from. Workspace patterns for nested documents/permissions.                    |
| <b>AppFlowy</b>          | <a href="https://github.com/AppFlowy-IO/AppFlowy">github.com/AppFlowy-IO/AppFlowy</a> | AGPL         | Learn from only — AGPL is poison for a SaaS.                                        |
| <b>Affine</b>            | <a href="https://github.com/toeverything/AFFiNE">github.com/toeverything/AFFiNE</a>   | MIT (core)   | Learn from. Block-based workspace inspiration.                                      |
| <b>Logseq</b>            | <a href="https://github.com/logseq/logseq">github.com/logseq/logseq</a>               | AGPL         | Learn from only.                                                                    |
| <b>TipTap</b>            | <a href="https://github.com/ueberdosis/tiptap">github.com/ueberdosis/tiptap</a>       | MIT          | Integrate. Working-paper editor foundation, ProseMirror-based, plays well with Yjs. |

| Project                  | URL                                                                                   | License            | Verdict                                                                          |
|--------------------------|---------------------------------------------------------------------------------------|--------------------|----------------------------------------------------------------------------------|
| <b>BlockNote</b>         | <a href="https://github.com/TypeCellOS/BlockNote">github.com/TypeCellOS/BlockNote</a> | MPL-2.0            | Integrate as the block-based narrative editor (Notion-like UX), built on TipTap. |
| <b>Plate.js</b>          | <a href="https://github.com/udecode/plate">github.com/udecode/plate</a>               | MIT                | Alternative to BlockNote — pick one. Recommend BlockNote for time-to-ship.       |
| <b>Yjs</b>               | <a href="https://github.com/yjs/yjs">github.com/yjs/yjs</a>                           | MIT                | Integrate. CRDT for collaborative editing.                                       |
| <b>TanStack Table</b>    | <a href="https://github.com/TanStack/table">github.com/TanStack/table</a>             | MIT                | Integrate. Working-paper grids and PBC list tables.                              |
| <b>AG Grid Community</b> | <a href="https://github.com/ag-grid/ag-grid">github.com/ag-grid/ag-grid</a>           | MIT<br>(community) | Use only if you outgrow TanStack — Enterprise tier license is meaningful cost.   |

B. Document AI / OCR / extraction

| Project | URL                                | License                                                     | Stars<br>(May 2026) | Verdict                                                                                                                                                                                                                                                                                                            |
|---------|------------------------------------|-------------------------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Docling | github.com/docling-project/docling | MIT (code),<br>Apache 2.0<br>(Granite-<br>Docling<br>model) | 59,593              | <b>Integrate —<br/>primary<br/>OCR/layout for<br/>narrative + tables.</b><br>Backed by IBM<br>Research, <a href="#">GitHub</a><br>LF AI & Data.<br><a href="#">GitHub</a> Supports<br>PDF, DOCX, PPTX,<br>XLSX, HTML,<br>audio. <a href="#">Docling</a><br><a href="#">GitHub</a><br>TableFormer<br>trained on 1M+ |

| Project              | URL                                                                                                                   | License                                   | Stars<br>(May 2026) | Verdict                                                                                                                                              |
|----------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      |                                                                                                                       |                                           |                     | tables.<br><div>ldp-software</div>                                                                                                                   |
| Granite-Docling-258M | <a href="https://huggingface.co/ibm-granite/granite-docling-258M">huggingface.co/ibm-granite/granite-docling-258M</a> | Apache 2.0                                | n/a                 | <b>Integrate</b> for on-prem/air-gapped deployments.<br>258M-param VLM<br><div>ldp-software</div> that "parses and processes documents in one shot." |
| Marker               | <a href="https://github.com/datalab-to/marker">github.com/datalab-to/marker</a>                                       | GPLv3 (code), modified AI Pubs Open Rail- | 30.9k               | <b>Avoid for SaaS.</b><br>Free for research, personal use, and startups under \$2M                                                                   |

| Project               | URL                                                    | License               | Stars<br>(May 2026) | Verdict                                                                                                                                                                    |
|-----------------------|--------------------------------------------------------|-----------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       |                                                        | M<br>(weights)        |                     | revenue/funding;<br><a href="#">GitHub</a> above<br>that, commercial<br>license required.                                                                                  |
| MinerU                | github.com/opencv/MinerU                               | AGPL                  | ~30k+               | Learn from only<br>(AGPL).                                                                                                                                                 |
| OpenDataLoader<br>PDF | github.com/opencv/opencv-<br>project/opencv-loader-pdf | Apache 2.0<br>(≥v2.0) | n/a                 | <b>Integrate as a<br/>deterministic<br/>fallback</b> for digital<br>PDFs. CPU-only,<br>rule-based, gives<br>bounding boxes for<br>every element.<br><a href="#">GitHub</a> |

| Project         | URL                                                                                                   | License     | Stars<br>(May<br>2026) | Verdict                                                         |
|-----------------|-------------------------------------------------------------------------------------------------------|-------------|------------------------|-----------------------------------------------------------------|
| Unstructured.io | <a href="https://github.com/Unstructured-IO/unstructured">github.com/Unstructured-IO/unstructured</a> | Apache 2.0  | ~12k                   | Optional fallback.                                              |
| LlamaParse      | cloud SaaS                                                                                            | proprietary | n/a                    | Use as API for complex narrative; pair with Docling for tables. |
| Reducto         | <a href="https://reducto.ai">reducto.ai</a>                                                           | proprietary | n/a                    | Optional cloud API for the hardest tables.                      |
| Nougat (Meta)   | <a href="https://github.com/facebookresearch/nougat">github.com/facebookresearch/nougat</a>           | CC-BY-NC    | ~9k                    | Skip for commercial.                                            |
| Tesseract       | <a href="https://github.com/tesseract-ocr/tesseract">github.com/tesseract-ocr/tesseract</a>           | Apache 2.0  | ~62k                   | Integrate as the OCR floor for                                  |

| Project           | URL                                                                                       | License    | Stars<br>(May<br>2026) | Verdict                                                                  |
|-------------------|-------------------------------------------------------------------------------------------|------------|------------------------|--------------------------------------------------------------------------|
|                   |                                                                                           |            |                        | scanned-image<br>PDFs.                                                   |
| <b>PaddleOCR</b>  | <a href="https://github.com/PaddlePaddle/PaddleOCR">github.com/PaddlePaddle/PaddleOCR</a> | Apache 2.0 | ~44k                   | Integrate for<br>Indian-language<br>OCR (V2 client<br>portal).           |
| <b>pdfplumber</b> | <a href="https://github.com/jsvine/pdfplumber">github.com/jsvine/pdfplumber</a>           | MIT        | ~7k                    | Integrate for<br>tabular extraction<br>from digital Indian<br>bank PDFs. |
| <b>Camelot</b>    | <a href="https://github.com/atlanhq/camelot">github.com/atlanhq/camelot</a>               | MIT        | ~3k                    | Integrate as a third<br>table-parser<br>fallback.                        |

| Project                       | URL                                                                                                 | License                   | Stars<br>(May 2026) | Verdict                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------|---------------------|---------------------------------------------------------------------------------------------------------|
| LayoutParser                  | <a href="https://github.com/Layout-Parser/layout-parser">github.com/Layout-Parser/layout-parser</a> | Apache 2.0                | ~5k                 | Learn from. Docling subsumes most use cases.                                                            |
| Firecrawl                     | <a href="https://github.com/mendableai/firecrawl">github.com/mendableai/firecrawl</a>               | AGPL (server),<br>MIT SDK | ~20k                | Use the cloud SDK (MIT) for MCA portal scraping; avoid forking the AGPL server.                         |
| Indian bank statement parsers | —                                                                                                   | —                         | —                   | <b>Build in-house.</b> No production-grade OSS parser exists for HDFC/ICICI/SBI PDFs. Competitive moat. |

## C. PDF and Excel parsing/rendering

| Project                  | License                                 | Verdict                                                               |
|--------------------------|-----------------------------------------|-----------------------------------------------------------------------|
| <b>PDF.js</b> (Mozilla)  | Apache 2.0                              | Integrate. Foundation for in-browser PDF viewer.                      |
| <b>pdf-lib</b>           | MIT                                     | Integrate. Add signed-tree-head footers to working-paper PDF exports. |
| <b>react-pdf</b>         | MIT                                     | Integrate. React wrapper over PDF.js.                                 |
| <b>pdfium</b>            | BSD-3                                   | Use server-side; avoid AGPL MuPDF.                                    |
| <b>SheetJS (xlsx)</b>    | Apache 2.0 community,<br>commercial Pro | Integrate community; evaluate Pro at scale.                           |
| <b>exceljs</b>           | MIT                                     | Integrate for write-side; SheetJS for read.                           |
| <b>openpyxl</b>          | MIT                                     | Integrate on Python workers.                                          |
| <b>react-spreadsheet</b> | MIT                                     | Integrate for the Excel pane in the click-to-source viewer.           |

D. PDF-Excel verification / reconciliation

No off-the-shelf OSS at production quality — **this is Orvyn's core IP**. Closest references: LangChain document loaders + custom comparison pipelines (Apache 2.0); pandas merge patterns; DataSnipper UX (closed; see Section K).

E. AI orchestration / agent frameworks

| Framework   | URL                             | License | Stars<br>(May 2026) | Verdict                                                                                                                                                                                                                                   |
|-------------|---------------------------------|---------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pydantic AI | github.com/pydantic/pydantic-ai | MIT     | 17,024              | <b>Primary for Verification Engine.</b><br>v1.0 shipped Sept 2025.<br>"FastAPI feeling applied to GenAI." <span>Speakeasy</span><br><span>GitHub</span> Scored 8/10<br>developer experience in the Nextbuild 90-day benchmark (highest of |

| Framework | URL                               | License | Stars<br>(May<br>2026) | Verdict                                                                                                                                                                       |
|-----------|-----------------------------------|---------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |                                   |         |                        | 5 frameworks tested);<br>caught 23 bugs at<br>compile time that would<br>have reached prod in<br>LangChain.                                                                   |
| LangGraph | github.com/langchain-ai/langgraph | MIT     | ~10k+                  | Use for V1.5+ workflows<br>needing durable<br>checkpointing and<br>human-in-the-loop<br>(PBC List Generator,<br>Variance Explanation).<br>v1.0 alpha shipped Sept<br>2, 2025. |
| LangChain | github.com/langchain-ai/langchain | MIT     | ~90k+                  | Use document loaders<br>only. Avoid LCEL.                                                                                                                                     |

| Framework                   | URL                                                                                                 | License   | Stars<br>(May<br>2026) | Verdict                                                                                                |
|-----------------------------|-----------------------------------------------------------------------------------------------------|-----------|------------------------|--------------------------------------------------------------------------------------------------------|
| <b>LlamaIndex</b>           | <a href="https://github.com/run-llama/llama_index">github.com/run-llama/llama_index</a>             | MIT       | ~35k+                  | Use only LlamaParse, not the agent layer.                                                              |
| <b>CrewAI</b>               | <a href="https://github.com/crewAIInc/crewAI">github.com/crewAIInc/crewAI</a>                       | MIT       | ~25k+                  | Skip.                                                                                                  |
| <b>AutoGen / AG2</b>        | <a href="https://github.com/microsoft/autogen">github.com/microsoft/autogen</a>                     | CC-BY-4.0 | ~32k+                  | Skip — near-zero security mechanisms, research-grade.                                                  |
| <b>DSPy</b>                 | <a href="https://github.com/stanfordnlp/dspy">github.com/stanfordnlp/dspy</a>                       | MIT       | ~20k+                  | <b>Use for prompt optimization</b> of the matcher agent (compile on synthetic, ship compiled prompts). |
| <b>Inspect AI (UK AISI)</b> | <a href="https://github.com/UKGovernmentBEIS/inspect_ai">github.com/UKGovernmentBEIS/inspect_ai</a> | MIT       | ~1k                    | <b>Integrate as the eval harness</b> for                                                               |

| Framework         | URL                            | License    | Stars<br>(May<br>2026) | Verdict                                                                               |
|-------------------|--------------------------------|------------|------------------------|---------------------------------------------------------------------------------------|
|                   |                                |            |                        | hallucination/regression tests.                                                       |
| BAML              | github.com/BoundaryML/baml     | Apache 2.0 | ~3k                    | Optional — Pydantic AI covers the same ground.                                        |
| Mastra            | github.com/mastra-ai/mastra    | MIT        | ~10k+                  | Skip — TS-first.                                                                      |
| Agent Squad (AWS) | github.com/awslabs/agent-squad | Apache 2.0 | ~3k                    | Skip — AWS coupling.                                                                  |
| Outlines          | github.com/dottxt-ai/outlines  | Apache 2.0 | ~10k                   | Integrate for forced structured-JSON output from local Llama 3.1 on the on-prem tier. |

## F. Cryptographic audit trail / immutable logs

| Project            | License     | Verdict                                                                                                                                            |
|--------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| @noble/ed25519     | MIT         | <b>Integrate.</b> Audited Ed25519 in pure JS, ~4 KB minified.                                                                                      |
| @noble/hashes      | MIT         | <b>Integrate</b> for SHA-256.                                                                                                                      |
| libsodium-wrappers | ISC         | Integrate on server side.                                                                                                                          |
| Trillian (Google)  | Apache 2.0  | Learn from; integrate only when we need our own production transparency log. Backs Sigstore Rekor v2 (Trillian-Tessera).<br><a href="#">GitHub</a> |
| Sigstore Rekor     | Apache 2.0  | Learn from. Reference architecture for our per-engagement Merkle log.                                                                              |
| merkletreejs       | MIT         | Integrate. Workhorse for the per-engagement Merkle tree.                                                                                           |
| AWS QLDB           | proprietary | Skip — QLDB is in EoL.                                                                                                                             |

| Project    | License    | Verdict                                                |
|------------|------------|--------------------------------------------------------|
| IPFS / CID | MIT/Apache | Learn from content-addressing patterns; do not deploy. |

## G. Multi-tenant SaaS infrastructure

| Project | License/Pricing                                                                                                                                                                                                                                                                                                  | Verdict                                                  |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| Clerk   | proprietary; \$25/mo Pro + \$0.02/MAU after 10k free; Clerk SSO is a paid add-on at \$100/mo                                                                                                                                                                                                                     | Growth-tier self-serve auth. Free up to 10k MAU.         |
| WorkOS  | proprietary; SSO \$125/connection/month for 1-15, \$100/mo for 16-30, \$80/mo for 31-50, \$65/mo for 51-100; Insign Directory Sync at same per-connection rates; Insign Audit Logs \$125/mo per SIEM connection + \$99/mo per million events stored; SuperTokens User Management free up to 1M MAU WorkOS Insign | Enterprise SSO/SCIM. ~\$2,000/mo for a 20-customer base. |

| Project                  | License/Pricing | Verdict                                                                                                                                                                        |
|--------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SuperTokens              | Apache 2.0      | Self-host fallback for paranoid customers / on-prem.                                                                                                                           |
| Auth0                    | proprietary     | Skip — expensive at scale; 300% B2C price hike (late 2023) hurt trust.                                                                                                         |
| NextAuth.js<br>(Auth.js) | ISC             | Use for the demo only; replace with Clerk/WorkOS in V1.                                                                                                                        |
| Drizzle ORM              | Apache 2.0      | <b>Primary ORM.</b> First-class RLS via <code>crudPolicy()</code> . <code>BuildMVPFast</code><br>~7.4 KB minified, zero deps, edge-native.                                     |
| Prisma                   | Apache 2.0      | Skip. v7 (Nov 2025) dropped Rust engine; v7.0–7.3 measurably slower than v6 under high concurrency.<br><code>BuildMVPFast</code> No built-in RLS.<br><code>BuildMVPFast</code> |

| Project     | License/Pricing           | Verdict                                                                                                                                                                                                                                                     |
|-------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kysely      | MIT                       | Use for complex audit queries where Drizzle is awkward.                                                                                                                                                                                                     |
| Trigger.dev | Apache 2.0                | Integrate for short async jobs in the demo.                                                                                                                                                                                                                 |
| Inngest     | Apache 2.0 / Cloud        | Alternative — pick Trigger.dev.                                                                                                                                                                                                                             |
| Temporal    | MIT                       | <b>Integrate for V1.5+ long-running orchestrations.</b> Durable execution, replay-from-history. (Startupik)                                                                                                                                                 |
| BullMQ      | MIT                       | Integrate for V1 Node queues.                                                                                                                                                                                                                               |
| LiteLLM     | MIT (OSS); Enterprise SKU | <b>Integrate as self-hosted gateway in Mumbai VPC.</b> Official catalog at <a href="https://models.litellm.ai">models.litellm.ai</a> lists <b>2,600+ models across 140+ providers</b> (Litellm) as of May 2026. PII masking via Presidio plugin, (Cloudron) |

| Project | License/Pricing | Verdict                                                             |
|---------|-----------------|---------------------------------------------------------------------|
|         |                 | virtual keys per engagement, spend tracking. <a href="#">GitHub</a> |

## H. PBC portal / file collection / client portal

No mature OSS Suralink alternative — build ourselves with: Resend / Postmark (email); Gupshup (WhatsApp BSP); Leegality (Aadhaar eSign, "a recognised electronic signature under the Second Schedule of the Information Technology Act, 2000"); DocuSign / Dropbox Sign (international fallback); Signzy (alternative). [Leegality](#)

## I. Workspace / collaboration

| Project    | License                     | Verdict                                 |
|------------|-----------------------------|-----------------------------------------|
| Liveblocks | proprietary (free <100 MAU) | Optional for V1.5 collaborative review. |
| Yjs        | MIT                         | Primary CRDT layer.                     |
| Automerger | MIT                         | Skip — Yjs is more mature in JS/TS.     |

| Project                | License | Verdict                                                |
|------------------------|---------|--------------------------------------------------------|
| <b>Linear</b> (closed) | —       | Study the keyboard-first UI; replicate cmdk + hotkeys. |
| <b>Affine</b>          | MIT     | Notion-clone UX inspiration.                           |

## J. Indian-specific tech

| Tool                                | License/Pricing                               | Verdict                                                                         |
|-------------------------------------|-----------------------------------------------|---------------------------------------------------------------------------------|
| <b>Tally Prime XML over HTTP</b>    | proprietary; HTTP server on default port 9000 | Build in-house adapter.                                                         |
| <b>tally-integration / Tally.Py</b> | MIT                                           | Learn from / fork; author warns of edge cases in production.<br><div>PyPI</div> |
| <b>CData Tally connector</b>        | proprietary, paid                             | Skip — license overhead.                                                        |
| <b>GST GSPs (Cygnet,</b>            | per-API                                       | <b>Mastergst</b> primary for V1 (transparent INR pricing, GSTR-                 |

| Tool                    | License/Pricing                                                                                                                | Verdict                                                   |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Mastergst,<br>Taxgenie) |                                                                                                                                | 1/3B/2A/2B/9/9C).                                         |
| MCA portal<br>scraping  | public                                                                                                                         | Scrape via Firecrawl/Spider;<br>cache aggressively.       |
| Razorpay                | per-tx                                                                                                                         | Primary for GST-compliant<br>invoicing.                   |
| Sarvam AI SDK           | INR pricing; ₹1,000 free credit; plans scale to<br>₹50,000; pay-per-use across<br>STT/TTS/translate/LLM (Tech Research Online) | V2 only — client portal<br>Hindi/regional languages.      |
| Krutrim AI SDK          | INR pricing                                                                                                                    | V2 fallback.                                              |
| Leegality               | per-document; free Basic plan                                                                                                  | <b>Primary e-sign.</b>                                    |
| NFRA inspection<br>data | public                                                                                                                         | Scrape; build NFRA Inspection<br>Readiness module on top. |

## K. Compliance / governance / audit-trail UI patterns (study, don't copy)

| Reference                                                                                                                                                                                                                  | What to study                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Vanta, Drata, Sprinto                                                                                                                                                                                                      | Evidence-collection automation UX. Use <b>Sprinto</b> (Indian) for our own SOC 2 / ISO 27001 readiness.                                                                                                                                                                                                                                      |
| <b>Optro</b> (formerly AuditBoard — rebranded as Optro on March 9, 2026 at the IIA's Great Audit Minds conference in Las Vegas; <b>Optro</b> CEO Raul Villar Jr.: "Today, AuditBoard becomes Optro")<br><b>PR Newswire</b> | Out-of-the-Box SOX RCM template; <b>AuditBoard</b> "update once, update everywhere" control linking; <b>AuditBoard</b> walkthrough → control → test linking; role-based dashboards with auto-PBC reminders; bidirectional linking between SOXHUB, OpsAudit, and ITRM. <b>Software Advice</b>                                                 |
| <b>Workiva</b>                                                                                                                                                                                                             | Cross-document linking for financial statement tie-out.                                                                                                                                                                                                                                                                                      |
| <b>DataSnipper</b>                                                                                                                                                                                                         | Excel-native tickmarks (Text / Sum / Validation / Exception / Table Snips). "Every snip, extraction, and AI output links back to its source document, creating a fully traceable, verifiable audit trail." <b>G2</b> Document Matching templates ("up to 94% efficiency gain"). <b>DataSnipper</b> Financial Statement Suite hover-to-source |

| Reference | What to study                                                                                       |
|-----------|-----------------------------------------------------------------------------------------------------|
|           | secondary view; tickmarks transfer to subsequent versions<br><div>DataSnipper</div> of the same FS. |

## L. Verification Engine libraries

| Library                                                | License     | Verdict                                                                                                              |
|--------------------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------|
| <b>rapidfuzz</b>                                       | MIT         | <b>Primary fuzzy matcher.</b> C++-backed, faster than fuzzywuzzy; <div>Plain English</div> MIT (vs. fuzzywuzzy GPL). |
| <b>thefuzz / fuzzywuzzy</b>                            | GPL         | Skip (license).                                                                                                      |
| <b>sentence-transformers</b>                           | Apache 2.0  | Integrate for offline embedding mode.                                                                                |
| <b>Gemini text-embedding-004</b><br>(Vertex AI Mumbai) | proprietary | Primary semantic-match embedding in cloud tier.                                                                      |
| <b>OpenAI text-embedding-3-small</b>                   | proprietary | Backup via LiteLLM.                                                                                                  |

| Library                        | License | Verdict                                                                |
|--------------------------------|---------|------------------------------------------------------------------------|
| react-pdf-highlighter          | MIT     | <b>Integrate.</b> 8.0.0-rc.0 is the current track. <a href="#">npm</a> |
| react-pdf-highlighter-extended | MIT     | Use if we need zoom + React 19. <a href="#">GitHub</a>                 |
| react-pdf-highlighter-plus     | MIT     | Optional — exports annotated PDFs natively. <a href="#">GitHub</a>     |

# PART 2 — CLAUDE CODE WORKFLOW OPTIMIZATION FOR 8 PEOPLE

Single biggest leverage point: **context discipline**. Community measurements indicate the "dumb zone" begins around 40% context fill; intelligence degrades materially past 300–400k tokens on the 1M model. For an 8-person team running 4–10 parallel sessions, this multiplies small mistakes unless you actively keep context small.

## 2.1 Monorepo vs polyrepo

**Decision: Turborepo monorepo.** Two SKUs share ~80% of code; CLAUDE.md is hierarchical;

Drizzle migrations + cross-package refactors are tractable. Nx is alternative; Turborepo wins on configuration overhead and Vercel integration.

## 2.2 CLAUDE.md structure

One root `CLAUDE.md` (~250 lines) plus per-package files. Root sections: (1) Repo overview + two-SKU model. (2) Tech stack reference. (3) **Domain glossary** — CARO 2020, SA 230, SQM 1, 3CD, Form 3CEB, GSTR-1/2A/2B/3B/9/9C, NFRA, ICAI, peer review, RPT, tickmark, working paper, PBC, walkthrough, RCM. (4) Coding conventions — TS strict, Drizzle schemas in `packages/db/schema/*.ts`, Zod in `packages/contracts/`, API responses `{data | error: {code, message}}`. (5) API design — REST + tRPC internal, gRPC across Node↔Python; cursor pagination; problem+json errors; `X-Tenant-Id` and `X-Engagement-Id` headers. (6) Testing — Vitest, Playwright, Inspect AI. (7) Anti-patterns — never bypass RLS with superuser, never log PII plaintext, never call LLMs without tenant tag, never store raw Aadhaar (only masked last-4). (8) Directory map. (9) Onboarding — `pnpm install && pnpm dev`. Per-package files at `apps/web`, `apps/api`, `services/ai`, `packages/verification`, `packages/audit-trail`.

## 2.3 Subagents

Per current Claude Code docs, subagents live in `.claude/agents/<name>.md` with YAML frontmatter; each gets a fresh context window, own tool allowlist, model, permissions. **Feature-**

**specific beats generic.** Ship these 5: `Mcp`

1. `verification-engine-extractor` — Read/Grep/Glob/Bash only; edits only `packages/verification/extractors/`. Loads Docling+rapidfuzz reference docs as a skill.
2. `drizzle-schema-evolver` — touches only `packages/db/`; runs `drizzle-kit generate` after every edit; forbids dropping columns.
3. `audit-trail-verifier` — read-only; runs Merkle integrity check; can call AWS CLI for S3 Object Lock state. Spawned daily by a hook.
4. `pbw-workflow-author` — focused on `packages/pbw/`. Loads Leegality API skill + WhatsApp template rules.
5. `security-reviewer` — `(disallowedTools: Write, Edit)`. Runs as a hook on every PR diff touching `auth/`, `crypto/`, `tenancy/`. `Claude`

## 2.4 MCP servers

Worth integrating (~5 servers / 58 tools costs ~55k tokens upfront, be deliberate): GitHub MCP, Linear MCP, Sentry MCP, Postgres MCP (read-only role; non-prod only via IAM), AWS MCP (describe-only), Notion MCP, Vercel MCP. **Do not install:** broad filesystem MCPs (redundant with built-ins), generic browser MCPs (Firecrawl SDK does the job cheaper). `Duet`

## 2.5 Slash commands and Skills

Skills under `.claude/skills/`: `verification-engine-spec`, `ca-domain-glossary` (auto-invokes on CARO/3CD/GSTR/SA/SQM/NFRA tokens), `audit-trail-conventions`, `tally-xml-formats`, `gst-data-schemas`, `working-paper-templates` (20-30 from CA-in-Residence), `dmdp-rules`. Slash commands: `/wp <topic>`, `/recon <pdf> <xlsx>`, `/migrate`, `/seed`, `/sth`.

## 2.6 Token optimization

Manual `/compact` at ~50% fill (auto-compact fires in the dumb zone). Subagents amortize tokens (20 reads + 12 greps stay in the subagent). Skills load lazily. Haiku for research subagents, Sonnet for implementation, Opus for hard reasoning. `Mcp`

## 2.7 Hooks and spec-driven flow

Hooks: `PreToolUse: Write|Edit` → `biome check`; `PostToolUse: Edit on packages/db/**` → `drizzle-kit check`; `PostToolUse: Edit on packages/verification/**` → 30-second pytest smoke; `Stop` → require commit message Linear ID. Spec-driven: spec in `specs/<feature>.md` → `/plan` → spawn `feature-implementer` subagent → `security-reviewer` runs on diff → human merge.

## 2.8 Pitfalls

20-step reasoning in main session instead of a subagent; generic subagents (collapse into main session); over-MCPing; edits in main session after research; trusting Claude on migrations without `drizzle-kit check`.

---

# PART 3 — DEMO TECH STACK (14-DAY BUILD)

## 3.1 Frontend

- **Next.js 15** (App Router, RSC) — final pick over Remix/Astro/SvelteKit.
- **React 19** — `use()`, Actions, `useFormStatus`. Use React 18.3 for packages depending on `react-pdf-highlighter`.
- **shadcn/ui + Tailwind v4** with the `@theme` directive — no `tailwind.config.ts` needed; CSS variables inline. shadcn beats Mantine/Chakra/Radix because we own the code. `DEV Community`
- **TanStack Query v5 + TanStack Table v8**.
- **Zustand** for ephemeral client state. Redux Toolkit rejected as overkill.
- **Recharts** for charts; **Tremor** on top for dashboard layouts.

- **React Hook Form + Zod** — shared with backend.
- **react-pdf-highlighter + react-spreadsheet** for click-to-source.
- **SheetJS, Framer Motion** judiciously, **lucide-react**.

### 3.2 Backend (demo)

Single Next.js app, API routes only. Postgres on **Neon** (`ap-south-1` Mumbai), Drizzle ORM. TS strict, Zod everywhere. Vercel KV / Upstash Redis for caches.

### 3.3 AI/ML for Screen 4 (Verification)

- **Gemini 2.0 Flash** via `@google/generative-ai` for planner + matcher.
- **Docling** on a separate Python Cloud Run service (Mumbai), invoked from a Next.js API route.
- **pdf-lib** for adding the Merkle-root footer.
- **SheetJS** for Excel; **rapidfuzz** (Python) for fuzzy matching; **Gemini text-embedding-004** for context-semantic.
- **Pydantic AI** for orchestration of the four agents inside the Python service.

ts

```
// packages/audit-trail/src/sign.ts
import * as ed from '@noble/ed25519';
import { sha256 } from '@noble/hashes/sha256';

export async function signEvent(privKey: Uint8Array, event: object) {
  const payload = canonicalize(event);
  const digest = sha256(new TextEncoder().encode(payload));
  const sig = await ed.signAsync(digest, privKey);
  return { payload, digest, sig };
}
```

A ~100-line Merkle tree (left-balanced, SHA-256), one tree per engagement, persisted to Postgres with daily Signed Tree Heads. Structurally identical to Sigstore Rekor's Trillian backing, only smaller. [GitHub](#)

### 3.5 Hosting (demo)

Vercel (frontend), Vercel Blob or S3 [ap-south-1](#) (files), Neon Postgres Mumbai, Cloud Run

Mumbai for the Docling/Pydantic AI service. Even the demo's Postgres and blobs should be in Mumbai — avoid Vercel KV outside India for tenant data.

### 3.6 Data for Acme Manufacturing — basis selection

**Pick V-Guard Industries Ltd. (FY24)** as the public-domain basis: ~₹4,857 crore consolidated revenue (+17.7% YoY), diversified product lines (Electronics, Electricals, Consumer Durables + Sunflame), multi-state plants (Kerala, Tamil Nadu, Himachal, Sikkim, Uttarakhand, Haryana), 36 branches with rich multi-state GST complexity. FY24 Annual Report at [vguard.in/uploads/investor\\_relations/V-Guard-AR-2023-24.pdf](https://vguard.in/uploads/investor_relations/V-Guard-AR-2023-24.pdf). Anonymize as "Acme Manufacturing Ltd." Full rules in Part 5. [Grokikipedia](#)

---

## PART 4 — V1 PRODUCTION TECH STACK (90-DAY BUILD)

### 4.1 Frontend (production)

Same component library. Multi-tenant routing via [app/\(tenant\)/\[orgSlug\]/...](#). WorkOS for Enterprise SSO/SCIM; Clerk for Growth. **Flagsmith self-hosted** in Mumbai VPC for flags. Sentry for errors; **PostHog self-hosted** for analytics (DPDP-friendly).



**NestJS** for Workspace API — modular architecture maps onto our 6 modules; built-in gRPC + microservices; class-validator + class-transformer; TS-first. **FastAPI** for AI services — Python is canonical for Docling, sentence-transformers, rapidfuzz, Pydantic AI; async model handles suspended inference well; Pydantic models shared verbatim. **Why two services:** cold-start cost of Docling/embedding models is unacceptable serverless; long-running jobs need persistent workers; DPDP residency is easier to audit with a discrete VPC. **gRPC** between services; REST for external; **NATS JetStream** for events. (EmporionSoft Pvt Ltd)

## 4.3 Database

**Aurora PostgreSQL 16** in `ap-south-1`; cross-region replica in `ap-south-2` (Hyderabad) for DR. **RDS Proxy** for pooling. Multi-tenancy via Drizzle + Postgres RLS, session variables (`SET LOCAL app.tenant_id = $tenant_id`). Drizzle Kit for migrations; Atlas backup. Continuous Aurora backtrack + daily snapshots + 7-year cold archive to Glacier Deep Archive.

## 4.4 Vector search

**pgvector** for V1; graduate to **Qdrant self-hosted** in Mumbai when corpus >2M chunks or recall drops.

## 4.5 Storage

**S3** `ap-south-1` with **Object Lock** in Compliance mode for 7-year retention. **MinIO** for on-prem. AES-256-GCM at rest with per-tenant DEKs wrapped by KMS keys.

## 4.6 Cache and queue

Redis (ElastiCache Mumbai); BullMQ for short jobs; **Temporal Cloud** (or self-hosted V1.5+) for long-running orchestrations.

## 4.7 Search

Postgres FTS with tsvector + GIN. Meilisearch when working-paper corpus >1M.

## 4.8 Real-time

WebSockets via NestJS Gateway. Yjs + y-websocket in V1.5.

## 4.9 Auth

WorkOS for Enterprise (SAML/OIDC/SCIM); Clerk for Growth self-serve. MFA mandatory for audit privileges.

## 4.10 LLM stack

**LiteLLM self-hosted in Mumbai VPC** — gives virtual keys per engagement, spend tracking per tenant, automatic failover, PII masking via Presidio plugin. Catalog at [models.litellm.ai](https://models.litellm.ai) lists **2,600+ models across 140+ providers** as of May 2026. **Default:** Gemini 2.0 Flash via Vertex AI Mumbai (zero-retention, lowest latency). **High-stakes:** Claude Sonnet 4.5 via AWS Bedrock with Geo CRIS APAC routing — Geo profiles "keep data processing within a specific geography (EU, US, APAC, etc.), ensuring compliance with data residency regulations." Note: Bedrock Claude availability in `ap-south-1` is partial — Vertex Gemini Mumbai is default; route to Bedrock APAC

for Claude. **On-prem (Year 2):** Llama 3.1 70B-Instruct or Sarvam-M on customer GPU via vLLM, routed via LiteLLM. **Caching:** LiteLLM Redis cache for deterministic prompts. [GitHub + 3](#)

## 4.11 Agent orchestration

**Verification Engine:** Pydantic AI runtime, four agents (planner/extractor/matcher/reviewer) as `Agent` instances with shared `RunContext` carrying `tenant_id`, `engagement_id`, working tape. Each returns a typed Pydantic model; outputs persist in Postgres for click-to-source. **PBC List Generator (V1.5):** LangGraph for durable checkpointing + human-in-the-loop pauses. **Variance Explanation (V1.5):** Pydantic AI compiled offline with DSPy on a held-out variance dataset.

## 4.12 Document processing pipeline

**Docling** (self-hosted `docling-serve`) default; **OpenDataLoader PDF** for deterministic digital PDFs; **LlamaParse** (cloud) for narrative where Docling underperforms (feature-flagged with consent); custom Python parsers in `services/ai/parsers/` for HDFC/ICICI/SBI/Axis and Tally XML. Pipeline orchestrated by the Pydantic AI extractor agent.

## 4.13 Cryptographic audit trail (production)

Ed25519 via libsodium server-side. One Merkle tree per engagement; leaves are SHA-256 of

canonicalized JSON events. Daily Signed Tree Heads to S3 Object Lock and gossiped to an independent log monitor (per Rekor's design — tamper-evident, with monitoring catching tampering). AWS CloudHSM for firm master keys on Enterprise tier; per-tenant DEKs wrapped by KMS. Export: signed PDF with the engagement's STH in the footer; offline verification possible. [GitHub](#)

## 4.14 Encryption

AES-256-GCM at rest; TLS 1.3 in transit; Customer-Managed Keys (CMK / BYO-KMS) for Enterprise; per-engagement DEKs rotated annually or on demand.

## 4.15 Security tooling and SOC 2

**Sprinto** (Indian) for compliance management; Snyk + Dependabot + Trivy for scanning; Tailscale for ops access (no public DBs); GuardDuty + Security Hub for AWS threats; Cloudflare for DDoS + WAF; AWS Secrets Manager (cloud) + Vault (on-prem).

## 4.16 Monitoring and observability

Sentry with sourcemap upload; Grafana Cloud + OpenTelemetry (metrics/logs/traces); Loki for log aggregation; LLM call logging via LiteLLM (tenant\_id, engagement\_id, model\_version, cost);

audit-trail dashboards for STH verification health and Merkle-root match rate.

## 4.17 DevOps / IaC

Terraform (not Pulumi — broader Indian hire-ability); Helm for on-prem; Docker Compose locally; **ECS Fargate** for production (avoid EKS until ~50 services); ECR with cosign image signing.

## 4.18 CI/CD

GitHub Actions. Vercel preview environments + ephemeral ECS task definitions for backend. Staging in Mumbai. Production blue-green via ECS deployment circuit breaker. DB migrations as a manually-approved CD step.

## 4.19 Testing

Vitest, Playwright, Stryker (only on `packages/verification/` and `packages/audit-trail/`), k6 for load, Inspect AI for agent regression evals.

## 4.20 Email / notifications

Resend (strict SPF/DKIM/DMARC); Gupshup as WhatsApp BSP (templates submitted 30 days

before launch); web push.

#### **4.21 E-sign**

Leegality primary (Aadhaar eSign per IT Act Schedule II); DocuSign fallback for international clients.

#### **4.22 Integration partners**

Tally Solutions (in-house XML/HTTP adapter); Mastergst (GST GSP) primary, Cygnet fallback; MCA via Firecrawl + Spider; banks: HDFC + ICICI + SBI in V1, Axis roadmap.

---

# PART 5 — DATA FOR THE DEMO

## 5.1 Public-domain mid-cap selection

| Company            | FY24<br>Revenue (₹<br>Cr)                                    | Plants                                                                                                | Multi-state<br>GST                                      | AR PDF                                                      | IR<br>dec |
|--------------------|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------------------|-----------|
| V-Guard Industries | ~4,857<br>(+17.7% YoY<br>consol.)<br><div>Grokikipedia</div> | Kerala, TN,<br>Himachal,<br>Uttarakhand,<br>Sikkim,<br>Haryana +<br>Sunflame<br>plants<br>(Faridabad) | Pan-India,<br>36<br>branches<br><div>Grokikipedia</div> | vguard.in/uploads/investor_relations/V-Guard-AR-2023-24.pdf | Yes       |
| Polycab India      | ~18,039<br>(FY24) /<br>28,884<br>(FY25)                      | 28 units<br>across 6<br>states<br><div>Wikipedia</div>                                                | Extensive                                               | polycab.com/ar2025                                          | Yes       |

| Company              | FY24<br>Revenue (₹<br>Cr) | Plants                                                        | Multi-state<br>GST | AR PDF          | IR<br>dec |
|----------------------|---------------------------|---------------------------------------------------------------|--------------------|-----------------|-----------|
| KEI<br>Industries    | ~8,104                    | Bhiwadi,<br>Chinchpada,<br>Pathredi,<br>Silvassa +<br>Gujarat | Yes                | BSE/NSE         | Yes       |
| Astral Ltd           | ~5,613                    | Gujarat, HP,<br>Rajasthan,<br>TN, Odisha                      | Yes                | astralpipes.com | Yes       |
| Cera<br>Sanitaryware | ~1,953                    | Two plants<br>Kadi Gujarat                                    | Limited            | cera-india.com  | Yes       |

| Company              | FY24<br>Revenue (₹<br>Cr)                                                         | Plants                           | Multi-state<br>GST | AR PDF               | IR<br>dec |
|----------------------|-----------------------------------------------------------------------------------|----------------------------------|--------------------|----------------------|-----------|
| Bajaj<br>Electricals | ~4,728<br><div>Equitymaster</div><br>/ 4,883<br>(FY25)<br><div>Equitymaster</div> | Chakan,<br>Wardha,<br>Ranjangaon | Yes                | bajajelectricals.com | Yes       |

**Recommendation: V-Guard Industries Ltd.** Best fits "mid-cap manufacturer, diversified, multi-state, ~₹3–8k cr." Astral is the runner-up if more SKU complexity is preferred. Polycab too large and politically sensitive (Dec 2023 IT raid + Mar 2024 Lockbit ransomware). Bajaj Electricals' FY24 affected by the Bajel Projects demerger. Cera too small. 

Wikipedia

5.2 Anonymization rules

(1) Replace "V-Guard" → "Acme Manufacturing Ltd." everywhere; rename subsidiaries. (2) Shift all dates by a deterministic offset (+127 days). (3) Re-base financial figures by a per-line random factor 0.85–1.15. (4) Substitute place names with fictional but geographically plausible Indian locations; keep state for GST realism. (5) Replace PANs, GSTINs, CIN with synthetic checksum-

valid values. (6) Replace director names with curated fictional list. (7) Replace bank account numbers / IFSC with synthetic values.

### **5.3 Tally XML**

Fork `tally-integration` (MIT) and generate: master export (groups, ledgers, stock items), 12 months of vouchers (sales, purchase, journal, receipt, payment, contra), opening balances, GSTIN tagging on each ledger.

### **5.4 GST filings**

Synthesize realistic GSTR-1, 2A, 2B, 3B, 9, 9C derived from the Tally voucher set; demo shows a real GSTR-9C reconciliation finding (e.g., 0.4% revenue variance between books and GSTR-1).

### **5.5 Bank statements**

HDFC, ICICI, SBI formats. 3 months each, ~600 transactions per bank. Include realistic noise: cheque clearing delays, NEFT failures, bank charges, GST on bank charges.

### **5.6 Working papers**

20-30 written by the CA-in-Residence: Bank Reconciliation, Cash Confirmation, Debtors Aging,

Inventory Cutoff, Revenue Cutoff, GST Reconciliation, RPT, CARO 2020 clause-by-clause, 3CD verification, Fixed Assets Movement, Provisions, Borrowings, Lease (Ind AS 116), Revenue Recognition (Ind AS 115).

## **5.7 Investor presentation for Screen 3**

12-page synthetic deck mirroring V-Guard's investor format — revenue by segment, EBITDA bridge, capex, working capital. Plant one deliberate inconsistency (chart says ₹4,902 cr, Excel says ₹4,857 cr) so the Verification Engine catches it on Screen 3.

## **5.8 Real vs. synthetic — recommendation**

**Synthetic for the demo; real consented sanitized data from a friendly CA contact for V1 pilots.** Real has irreproducible texture (the way Indian SMEs actually book JEs, typos, legacy CoA cruft) but legal overhead in 14 days is prohibitive.

---

## 6.1 Pipeline

Input: PDF (50p) + Excel (20 sheets)



Planner Agent (Pydantic AI on Gemini 2.0 Flash)



├─> Extractor Agent (Docling + custom parsers)

├─> Matcher Agent (rapidfuzz + embeddings)

└─> Reviewer Agent (Claude Sonnet via Bedrock APAC)



Output: Verification Report (JSON) + source positions + audit-trail entry

## 6.2 OCR / extraction

**Default: Docling** with the `granite-docling` VLM (Apache 2.0). Returns typed

`DoclingDocument` with bounding boxes — critical for click-to-source. Fallback to OpenDataLoader PDF for digital-native; Tesseract + PaddleOCR for scanned-image. Indian bank statements: custom Python parsers per bank — the moat.

## 6.3 Context extraction

For each extracted number, capture column header(s), row label, table title, section heading, page, bbox, surrounding text:

```
python
```

```

class Claim(BaseModel):
    value: Decimal
    unit: Literal['INR', 'INR_lakh', 'INR_crore', 'percent', 'count']
    sign: Literal['+', '-']
    page: int
    bbox: BBox
    column_headers: list[str]
    row_label: str
    table_title: Optional[str]
    section_heading: Optional[str]
    surrounding_text: str
    source_doc: str

```

Indian-format handling: parse "5,42,30,000", "(2.5)" as -2.5, "Rs. 45.2 cr" → 452,000,000, lakhs vs. crore, "Net of GST" annotations.

## 6.4 Matching algorithms

**Exact** after normalization. **Fuzzy** via rapidfuzz (token\_set\_ratio labels, partial\_ratio substrings). **Context-semantic** via Gemini text-embedding-004 (embed <row\_label> ·

`<column_headers> · <table_title>`]; cosine  $\geq 0.88$ ). **Tolerance:** per-engagement, default ₹5,000 or 0.1%, whichever greater. Rupee→paisa round-trip.

## 6.5 Agent orchestration

Each agent a Pydantic AI `Agent`. Shared `RunContext` carries tenant, engagement, doc-set, cumulative reasoning log. Planner emits plan; extractor returns claims; matcher returns matches with scores; reviewer reads matches + context and confirms or downgrades. All four replayable deterministically from persisted intermediate state.

## 6.6 Confidence scoring

```
confidence = w1·exactness + w2·fuzzy_score + w3·semantic_sim  
            + w4·reviewer_agreement - w5·domain_penalty
```

Three bands: **High**  $\geq 0.92$  (auto-tick), **Medium** 0.75–0.92 (human confirm), **Low**  $< 0.75$  (exception). Rejections feed back as DSPy training signal.

## 6.7 Click-to-source viewer

react-pdf-highlighter (left) + react-spreadsheet (right). Clicking a verified pair scrolls both panes

to the source with synchronized highlights — the DataSnipper pattern: every output links back to its source for a fully traceable, verifiable audit trail. (G2)

## 6.8 Performance targets

50-page PDF + 20-sheet Excel in  $\leq 5$  minutes on a single FastAPI worker with 4 vCPU + 16 GB. Docling page-batch parallelism (batch=8); rapidfuzz (cdist) for bulk similarity; embedding cache for repeated row labels; reviewer agent only on Medium-band matches (~5-15%), capping LLM cost.

## 6.9 Multi-tenancy

Each verification job runs in a separate (engagement\_id)-scoped worker. Blobs decrypted with per-engagement DEK only inside the worker; results encrypted back to Postgres. RLS prevents cross-engagement leakage on a query bug.

## 6.10 Cost per verification

Docling on Cloud Run T4 ~₹3; planner (Gemini Flash, ~5k tokens) ~₹0.20; extractor ~zero LLM; matcher embedding ~₹0.05; reviewer on Claude Sonnet ~₹8. **Total ~₹12 per verification.** At Aditya Birla volumes (5-10 days  $\rightarrow$  <1 hour), unit economics 1000:1.

---

# PART 7 — MULTI-TENANCY AND DATA ISOLATION

## 7.1 RLS implementation

```
sql

ALTER TABLE working_papers ENABLE ROW LEVEL SECURITY;
CREATE POLICY tenant_isolation ON working_papers
  USING (tenant_id = current_setting('app.tenant_id')::uuid);
```

Drizzle's `crudPolicy()` generates these; NestJS middleware sets `SET LOCAL app.tenant_id = ${ctx.tenantId}`. **Enterprise:** separate Aurora cluster per tenant via Terraform module `enterprise-tenant`.

## 7.2 Per-tenant encryption keys

Tenant master key in KMS; per-engagement DEKs wrapped under it. DEKs cached in worker memory for the lifetime of a verification job, then zeroed.

CloudHSM-backed or customer-account KMS via cross-account roles. Merkle-log signing keys can be customer-held — we never see them.

## 7.4 VPC peering

Single-tenant VPC peered with customer's internal network for paranoid customers. On-prem Helm chart is the alternative.

## 7.5 On-prem deployment

Helm chart `orvyn/orvyn`: Postgres (or external), MinIO, vLLM-served Llama 3.1 70B or Sarvam-M, local LiteLLM gateway, no outbound network. Tested on EKS, OpenShift, bare-metal Kubeadm.

---

## PART 8 — DEPLOYMENT AND DEVOPS

### 8.1 AWS account structure

AWS Org: `orvyn-prod` (Mumbai + Hyderabad), `orvyn-staging`, `orvyn-dev`, `orvyn-security` (centralized GuardDuty/CloudTrail/Security Hub), `orvyn-shared` (ECR/KMS).

### 8.2 VPC

Public, private, DB subnets across 3 AZs in `ap-south-1`. NAT gateway in each AZ. Private subnets for ECS; DB subnets isolated. VPC endpoints for S3, KMS, Secrets Manager, ECR.

### 8.3 Compute

ECS **Fargate** over EKS until we cross ~50 services. Auto-scales on CPU + custom CloudWatch queue-depth metrics.

### 8.4 Aurora

PostgreSQL 16, one writer + two readers across AZs. Reader endpoint for analytics. Cross-region replica in `ap-south-2` for DR. RTO 1h, RPO 5m via PITR.

## 8.5 S3, CloudFront, Route53, ACM, WAF

S3 with policies denying non-TLS and non-VPC for sensitive buckets; Object Lock for audit-trail bucket. CloudFront for static assets only. Route53 + ACM. **Cloudflare WAF** in front of CloudFront (better rule library + bot management).

## 8.6 Secrets, KMS, IAM

All secrets in Secrets Manager; rotated automatically. KMS keys per env + per tenant (Enterprise). IAM least-privilege; no (\*) resources outside bootstrap role. Tagging: `env`, `service`, `owner`, `cost-center` on every resource.

## 8.7 Cost monitoring

AWS Cost Explorer + Cost Anomaly Detection. Per-tenant attribution via tag propagation.

**Approximate AWS cost projections (Mumbai, list price, no commitment discounts — extrapolations, expect 20-30% variance)**

| <b>Tier</b>                                | <b>10 customers</b> | <b>100 customers</b>  | <b>1,000 customers</b> |
|--------------------------------------------|---------------------|-----------------------|------------------------|
| Aurora (writer + 2 readers, db.r6g.xlarge) | ~\$1,800/mo         | ~\$5,400/mo (sharded) | ~\$24,000/mo           |
| ECS Fargate (web + api + workers)          | ~\$900              | ~\$4,500              | ~\$22,000              |
| S3 + Object Lock (1 TB → 200 TB)           | ~\$40               | ~\$2,000              | ~\$15,000              |
| Bedrock Claude Sonnet (Geo APAC)           | ~\$300              | ~\$3,000              | ~\$25,000              |
| Vertex AI Gemini Flash                     | ~\$150              | ~\$1,500              | ~\$12,000              |
| CloudWatch + GuardDuty + SH                | ~\$200              | ~\$1,000              | ~\$6,000               |
| Data transfer                              | ~\$100              | ~\$800                | ~\$5,000               |
| <b>Total infra</b>                         | <b>~\$3,500</b>     | <b>~\$18,000</b>      | <b>~\$110,000</b>      |
| Per-customer infra                         | ~\$350              | ~\$180                | ~\$110                 |

SaaS on top: Sentry ~\$50/\$300/\$1,500, Vercel ~\$20/\$200/\$2,000, **WorkOS** ~\$1,250/\$8,000/\$70,000 (per-connection scaling: \$125/conn for 1-15, \$100/conn for 16-30, \$80/conn for 31-50, \$65/conn for 51-100), Linear/Notion ~\$200/\$2,000/\$15,000. (Infisign)

---

## PART 9 — OBSERVABILITY AND SECURITY MONITORING

### 9.1 Errors / logs / metrics / traces

Sentry with sourcemap upload on every deploy; tenant + engagement IDs always tagged. OpenTelemetry auto-instrumentation in NestJS + manual spans on verification pipeline stages. Logs structured JSON → CloudWatch Logs → Loki. Prometheus exporters → Grafana Cloud. OTel collector → Grafana Tempo. PagerDuty alerts tiered: P1 (signature mismatch, RLS bypass attempt, audit-log gap), P2 (verification SLA breach), P3 (cost anomaly).

### 9.2 LLM-specific observability

LiteLLM logs every prompt + response to Postgres (`llm_calls`) with `tenant_id`, `engagement_id`, `agent_name`, `model`, `input/output tokens`, `cost_inr`, `latency_ms`, `request_hash`. Dashboards: cost

per engagement, reviewer-downgrade rate (hallucination-prone prompts), token efficiency over time.

### 9.3 Audit-trail-specific observability

Daily STH verification job; Merkle proof-of-inclusion endpoint for independent auditor verification; alarms on signature failures, STH generation failures, S3 Object Lock policy drift.

### 9.4 Security monitoring

GuardDuty + Security Hub. Tenant access logged into the per-engagement Merkle tree. WorkOS Audit Logs + Clerk events → CloudWatch. WAF rate rules + custom CloudWatch alarms on `requests_per_tenant_per_minute`. Bucket policies deny large downloads; CloudTrail alarms on >100 MB S3 GET from non-VPC. SOC 2 evidence collection automated through Sprinto.

---

## PART 10 — TECH GAPS (areas previously under-specified)

- **API design philosophy:** REST for public, tRPC for internal Next.js↔Next.js, gRPC across Node↔Python. URI versioning (`/v1/`); 18-month sunset minimum. Cursor pagination;

problem+json errors; `X-RateLimit-*`; idempotency keys on every state-changing endpoint.

- **Frontend state at scale:** TanStack Query (server), Zustand (ephemeral), URL params (shareable filters). No global Redux.
- **CARO 21 / 3CD 44:** `<ClauseForm>` component per variant with conditional sub-clauses + Zod validation. CARO 2020 has 21 clauses; today partner sign-off takes 12 hours — V1.5 cuts to 2 hours.
- **i18n:** English V1. Hindi/Tamil/Marathi/Bengali for client portal in Y2 via **Sarvam-Translate** (22 Indian languages, open weights) + next-intl ICU MessageFormat.
- **Accessibility:** WCAG 2.1 AA mandatory — enterprise procurement requires it. shadcn + Radix gives ~80% free; axe-core in Playwright.
- **Performance budgets:** TTI  $\leq 3$ s on 4G (Tier-1 city); LCP  $\leq 2.5$ s; INP  $\leq 200$ ms. Largest pages chunked with React.lazy.
- **Offline / unreliable internet:** drafts auto-save to IndexedDB; Yjs CRDT in V1.5 for conflict resolution.
- **Mobile:** partner check-status on phones — `/portfolio` route fully responsive in V1.
- **Print stylesheets:** `@media print` on every working paper; partner-grade PDF via Puppeteer workers.
- **Search:** Postgres FTS V1, Meilisearch >1M docs.

- **Email deliverability:** Resend; strict SPF/DKIM/DMARC; dedicated subdomain `mail.orvyn.in`.
- **WhatsApp BSP:** Gupshup. Templates submitted 30 days pre-launch — UTILITY for PBC reminders, AUTHENTICATION for OTP.
- **E-sign:** Leegality primary; Aadhaar eSign was given legal sanctity via Gazette Notification GSR 61(E) dated January 28, 2015 ("Electronic Signature or Electronic Authentication Technique and Procedure Rules, 2015"). Aadhaar OTP path for retail; biometric fallback. `Leegality`
- **GST invoicing:** Razorpay with GST automation; auto-IRN via GSP.
- **Refunds / trials:** 14-day pilot SOW (not self-serve — ₹40L+ deals).
- **API rate limiting per tenant:** Redis token bucket; default 600 rpm, configurable.
- **DDoS:** Cloudflare + AWS Shield Standard. Shield Advanced (~\$3k/mo) only after first attack or first enterprise demand.
- **Backup encryption / offsite:** cross-region replication of audit-trail bucket with separate KMS key; quarterly restore drills.
- **DR runbooks:** Notion page per failure mode (DB lost, region down, KMS compromised, audit-log breach).
- **OSS license scanning:** FOSSA or Snyk License.
- **Container image security:** Trivy in CI; reject on HIGH unfixed.

- **Dependency updates:** Renovate weekly; auto-merge minor/patch with passing tests.
- **Browser matrix:** Chrome  $\geq 120$ , Edge  $\geq 120$ , Safari  $\geq 17$ , Firefox  $\geq 120$ . No IE.
- **Partner laptop story:** full disk encryption + MDM-enforced; customer data only via VPN; offline working papers encrypted in IndexedDB with per-session key.
- **CRDTs for offline edits:** Yjs (V1.5).
- **Pre-mortem at scales:**
  - **10 customers:** NestJS monolith fine; verification SLA is the failure mode — invest in worker autoscaling.
  - **100 customers:** Postgres connection limits bite — RDS Proxy + per-tenant pool. LLM cost dominates infra; LiteLLM caching critical. Mumbai-based on-call rota.
  - **1,000 customers:** shard tenants across multiple Aurora clusters by tenant-hash. Move queue BullMQ  $\rightarrow$  Temporal entirely. Vector store on Qdrant. S3 Object Lock + Glacier tiering essential.
- **Cost per customer:**  $\sim \$350$  at 10  $\rightarrow \sim \$180$  at 100  $\rightarrow \sim \$110$  at 1,000.
- **Productivity tools:** Linear, Notion, Slack, Loom, GitHub Projects.
- **Documentation:** dev docs in `docs/dev/` (Nextra); user docs `docs/user/`; admin guides `docs/admin/`.
- **Customer support:** Crisp for chat (cheap, Indian-friendly); roll-your-own ticketing on Linear.

- **Status page:** Better Stack.
  - **Bug bounty:** private Bugcrowd 6 months post-V1; public 12 months.
  - **Pentest:** annual external (NII / SISA — both Indian) — required for SOC 2 Type II.
  - **Insurance:** cyber liability  $\geq$  ₹10 cr; professional indemnity / E&O  $\geq$  ₹5 cr — required by enterprise procurement.
  - **Legal:** MSA, DPA (DPDP-aligned), SCC for cross-border, ICAI advertising-rule-compliant marketing.
  - **Engineering culture:** required code review by one peer + security-reviewer subagent; weekly blameless post-mortems; paid on-call rotation.
- 

## PART 11 — PROJECT STRUCTURE

```
orvyn/  
├─ apps/  
│   ├─ web/           # Next.js 15 frontend  
│   ├─ api/           # NestJS workspace API  
│   └─ docs/          # Nextra  
└─ services/
```

```
|   ├── ai/                # FastAPI + Pydantic AI + Docling
|   ├── parsers/          # Indian bank PDF parsers
|   └── orchestrator/     # Temporal worker (V1.5+)
└── packages/
    ├── ui/               # shadcn + Tailwind primitives
    ├── db/               # Drizzle schema + migrations
    ├── contracts/        # Zod + Pydantic shared schemas
    ├── audit-trail/       # Ed25519 + Merkle library
    ├── verification/     # Core matching algos (TS bindings)
    ├── pbc/
    ├── tally/
    └── gst/
└── infra/
    ├── terraform/
    ├── helm/
    └── docker-compose.yml
└── .claude/
    ├── agents/
    ├── skills/
    ├── hooks.json
    └── CLAUDE.md (root)
└── turbo.json
```

└─ package.json  
└─ README.md

**Tooling:** Husky + lint-staged; Biome for JS/TS (faster than ESLint + Prettier); ruff + mypy + pytest for Python; Volta for Node version pinning; Dockerfile per service with multi-stage builds, distroless final images.

---

## PART 12 — DEVELOPMENT TIMELINE

### 12.1 Days 1-14 (Demo)

| Day | Output                                                                                       | Lead   |
|-----|----------------------------------------------------------------------------------------------|--------|
| 1   | Repo bootstrap; Turborepo + Next.js + Drizzle + Neon; CLAUDE.md root + per-package skeletons | Tejash |
| 2   | shadcn + Tailwind v4 + design tokens; Screen 1 (Dashboard) shell                             | F2     |

| Day | Output                                                                       | Lead                    |
|-----|------------------------------------------------------------------------------|-------------------------|
| 3   | Acme Manufacturing seed data v1 (Tally XML + bank PDFs from V-Guard FY24)    | Kedar + CA-in-Residence |
| 4   | Screen 2 (Engagement Workspace)                                              | F2 + F3                 |
| 5   | FastAPI service skeleton; Docling installed; first PDF parsed                | Kedar                   |
| 6   | rapidfuzz matcher prototype; Pydantic AI planner agent on Gemini Flash       | Kedar                   |
| 7   | Click-to-source viewer prototype (react-pdf-highlighter + react-spreadsheet) | F4                      |
| 8   | Screen 3 (Investor Deck vs Excel) end-to-end                                 | Kedar + F4              |
| 9   | Audit trail real (Ed25519 + Merkle); PDF export footer                       | Tejash                  |
| 10  | Screen 4 (PBC Portal) sketch                                                 | F5                      |

| Day                                                                                                  | Output                                                           | Lead            |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|-----------------|
| 11                                                                                                   | Screen 5 (Cash WP) and Screen 6 (Revenue/GST)                    | F2 + CA-in-Res. |
| 12                                                                                                   | Screen 7 (Review + Audit Trail Export) — signed PDF, JSON proof  | Tejash          |
| 13                                                                                                   | Polish, demo script, video record, stress-test on 5 engagements  | All             |
| 14                                                                                                   | Bake day, fix top-10 issues, deploy to Vercel + Cloud Run Mumbai | All             |
| <b>Risk: Docling Cloud Run cold-starts</b><br>— mitigate with min-instances $\geq 1$ + cron warming. |                                                                  |                 |

# 12.2 Days 15-105 (V1)

| Week | Module                                                                         |
|------|--------------------------------------------------------------------------------|
| 3    | Auth (Clerk + WorkOS), multi-tenant scaffolding with RLS                       |
| 4    | Engagement Workspace V1 (CRUD, permissions, audit log)                         |
| 5    | Verification Engine V1 — hardened pipeline, autoscaling workers (hero feature) |
| 6    | PBC Portal V1 — Leegality integration                                          |
| 7    | Document Ingestion — Tally adapter, 3 bank parsers, manual upload              |
| 8    | Cash Working Paper module                                                      |
| 9    | Revenue/GST Working Paper module                                               |

| Week                                                                   | Module                                                              |
|------------------------------------------------------------------------|---------------------------------------------------------------------|
| 10                                                                     | Review + Audit Trail Export V1                                      |
| 11                                                                     | <b>Pilot 1 onboarding</b> — friendly CA firm                        |
| 12                                                                     | Hardening: SOC 2 controls, Sprinto setup                            |
| 13                                                                     | <b>Pilot 2 onboarding</b> — internal audit team (Aditya Birla port) |
| 14                                                                     | Performance — verification $\leq 5$ min under load                  |
| 15                                                                     | <b>Pilot 3 onboarding</b> — Dubai-based Indian firm                 |
| Day 91 (~Week 13): <b>Conversion week</b> — pilot<br>1 signs ₹40L AMC. |                                                                     |
|                                                                        |                                                                     |

# PART 13 — POST-V1 ROADMAP (LIGHT)

## V1.5 (months 4-6)

- **CARO 2020 module** — 21-clause workflow; partner sign-off 12h → 2h.
- **Form 3CD** — 44-clause tax audit; integrates Tally + GSTR.
- **GSTR-9C full reconciliation** — books vs. GSTR-1 vs. 3B.
- **NFRA Inspection Readiness** — dashboards mirroring NFRA checklists.
- **PBC List Generator** — LangGraph-orchestrated; partner reviews; dispatches via WhatsApp + portal.
- **Variance Explanation Engine** — Pydantic AI agent drafts variance commentary from GL.

## V2 (months 7-12)

- **Anomaly detection** — Benford, Z-score, isolation forests + LLM patterns.
- **Working paper draft generator** — end-to-end working paper with tickmarks pre-applied.
- **Independence / conflict resolution engine** — ICAI conflict automation.
- **Multi-LLM routing** via LiteLLM + DSPy-optimised model selection.
- **On-prem deployment** — Helm hardened; Llama 3.1 70B / Sarvam-M on customer GPU.

- **ISO 42001** (AI management) readiness — auditable AI policies, model risk register, dataset lineage in Postgres.

## **Recommendations (Staged)**

**Now (Days 1-14):** Lock the stack to Next.js 15 + Drizzle + Neon + Docling + Pydantic AI + Gemini Flash + @noble/ed25519. Resist any temptation to add Temporal, NestJS, or LiteLLM to the demo — they belong in V1.

**Day 14 → Day 30:** Split the monolith. NestJS workspace API + FastAPI AI service. WorkOS contract signed. AWS Mumbai accounts provisioned. Sprinto onboarding starts.

**Day 30 → Day 60:** Verification Engine hardened. Bank parsers for HDFC + ICICI + SBI in production. Tally adapter v1. PBC + Leegality flow live.

**Day 60 → Day 91:** Pilots onboard one per week. Each adds one new working paper template and one new edge case to the test suite.

**Day 91:** First ₹40L conversion. Fund Sprinto + WorkOS + Bedrock + Vertex commitments for 12 months.

**Triggers that change the plan:**

- Verification SLA misses (>5 min on p95) any time → invest in worker autoscaling and Docling batch tuning before any new module.
- LLM cost per engagement >₹50 → audit prompts, increase caching, run a DSPy compilation pass.
- Pilot demands on-prem before Year 2 → fork-lift Year-2 plan: Llama on their GPU, Helm chart, MinIO. Charge accordingly.
- DPDP enforcement begins earlier than 13 May 2027 → tighten retention defaults, accelerate consent-manager integration.

## Coverage Table

| Plan item                                | Covered in                                                                                 |
|------------------------------------------|--------------------------------------------------------------------------------------------|
| 1. OSS landscape (12 sub-categories A-L) | Part 1  |
| 2. Claude Code workflow                  | Part 2  |
| 3. Demo tech stack (14-day)              | Part 3  |
| 4. V1 production stack (90-day)          | Part 4  |

| Plan item                                    | Covered in                                                                                                  |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| 5. Acme Manufacturing dataset                | Part 5                   |
| 6. Verification Engine deep architecture     | Part 6                   |
| 7. Multi-tenancy and isolation               | Part 7                   |
| 8. Deployment / DevOps (AWS Mumbai)          | Part 8                   |
| 9. Observability & security monitoring       | Part 9                   |
| 10. Tech gaps                                | Part 10                  |
| 11. Project structure + CLAUDE.md            | Part 11 + §2.2           |
| 12. Day 1-14 + Day 15-105 timeline           | Part 12                  |
| 13. V1.5 + V2 roadmap                        | Part 13                  |
| Cost projections at 10/100/1000 customers    | Part 8                   |
| Indian regulatory context (DPDP, NFRA, ICAI) | Parts 4, 6, 10, Caveats  |

| Plan item       | Covered in                                                                                          |
|-----------------|-----------------------------------------------------------------------------------------------------|
| Recommendations | Recommendations  |
| Caveats         | Caveats          |

## Caveats

- **AuditBoard rebranding:** AuditBoard rebranded as Optro on March 9, 2026 at the IIA's Great Audit Minds conference in Las Vegas (CEO Raul Villar Jr.: "Today, AuditBoard becomes Optro"). Modules (SOXHUB, OpsAudit, CrossComply, ITRM, TPRM) remain in use under the Optro umbrella.
- **Marker licensing:** 30.9k stars on the canonical datalab-to/marker repo as of May 2026; modified AI Pubs Open Rail-M weights are free only for research, personal use, and startups under \$2M revenue/funding. Plan for licensing or Docling migration before Orvyn raises a meaningful round.
- **Bedrock Claude availability in Mumbai:** partial in `ap-south-1`, changes monthly. Default to Vertex AI Gemini Mumbai; route to Bedrock APAC Geo CRIS for Claude (Geo profiles route within a geography but not necessarily inside `ap-south-1`).

- **Sarvam pricing:** INR-denominated; pay-per-use plans scale from ₹0 to ₹50,000 with ₹1,000 starting credit; bulk enterprise pricing via [developer@sarvam.ai](mailto:developer@sarvam.ai).
- **Speculative claims flagged:** AWS unit-cost projections at 1,000 customers are extrapolations using publicly observed Aurora and Bedrock list prices and assume no committed-use discounts; expect 20–30% variance at scale.
- **DPDP enforcement timeline:** substantive obligations not enforceable until 13 May 2027 per current MeitY notification of 13 November 2025; could be brought forward by amendment.
- **Polycab risk:** if ever reconsidered as the dataset basis, note December 2023 IT Department raid and March 2024 Lockbit ransomware — interesting audit-risk narrative but distracting from a demo. V-Guard avoids these issues.
- **Indian listed company financials:** figures cited (V-Guard ~₹4,857 cr, KEI ~₹8,104 cr, Astral ~₹5,613 cr, Cera ~₹1,953 cr, Bajaj Electricals ~₹4,728 cr) are FY24 from publicly available annual reports / Equitymaster / Screener; verify against the original ARs before any external use.
- **LiteLLM coverage:** the GitHub README headline of "100+ LLM APIs" is a marketing minimum; the live model catalog at [models.litellm.ai](https://models.litellm.ai) lists 2,600+ models across 140+ providers as of May 2026.
- **Star counts:** Docling at 59,593 and Pydantic AI at 17,024 are as of May 12, 2026 from the respective official GitHub organisations — these are fast-moving repos; verify live counts

before external citation.

- **Aadhaar eSign legal basis:** Aadhaar eSign was given legal sanctity via Gazette Notification GSR 61(E) dated January 28, 2015 ("Electronic Signature or Electronic Authentication Technique and Procedure Rules, 2015"), making it a recognised electronic signature under the Second Schedule of the IT Act, 2000.